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## Probing MiniBooNE Data through Neural Nets and Dimensionality Reduction

This document provides a background, our previous ML-based analyses, and proposed next steps for investigating the MiniBooNE Low-Energy Excess (LEE) through dimensionality reduction (DR) techniques.

### Background

The MiniBooNE experiment produces a muon neutrino beam through proton collisions on a beryllium target, generating pions that decay primarily via  $\pi^+ \rightarrow \mu^+ + \nu_\mu$ . The purpose of the MiniBooNE detector is to detect electron neutrinos that may arise from neutrino oscillations, particularly through charged-current quasi-elastic (CCQE) interactions:  $\nu_e + n \rightarrow e^- + p$ . However, this signal must be distinguished from large backgrounds. The most significant of these backgrounds are delta particles produced through various muon neutrino reactions, which decay to produce either a photon directly or a  $\pi^0$  particle that typically decays to a photon pair. These photons eventually pair produce as electron-positron pairs in the mineral oil within the detector, and these resulting leptons then undergo bremsstrahlung and other processes to produce electron-positron cascades within the detector. In short, the experiment features large backgrounds that must be meticulously understood in order to place trust in event classification.

The MiniBooNE experiment has reported quantities of neutrino oscillations which are incompatible with Standard Model predictions. This so-called Low-Energy Excess (LEE) of neutrino oscillations from muon neutrinos to electron neutrinos is specifically in the sub-275 MeV range, with a  $4.7\sigma$  anomaly corresponding to approximately 460 unexplained electron-like events. In previous analysis, Boosted Decision Trees (BDT) have been trained on simulated data for event classification. An analysis based on selection cuts of nine reconstructed variables (energy, vertex, likelihood ratios, etc.) has also been implemented and published. These two methods do not discriminate data based on the location of individual events and clusters thereof in high-dimensionality feature space (rather, they implement cuts on *all events* without regard for the clustering of the data).

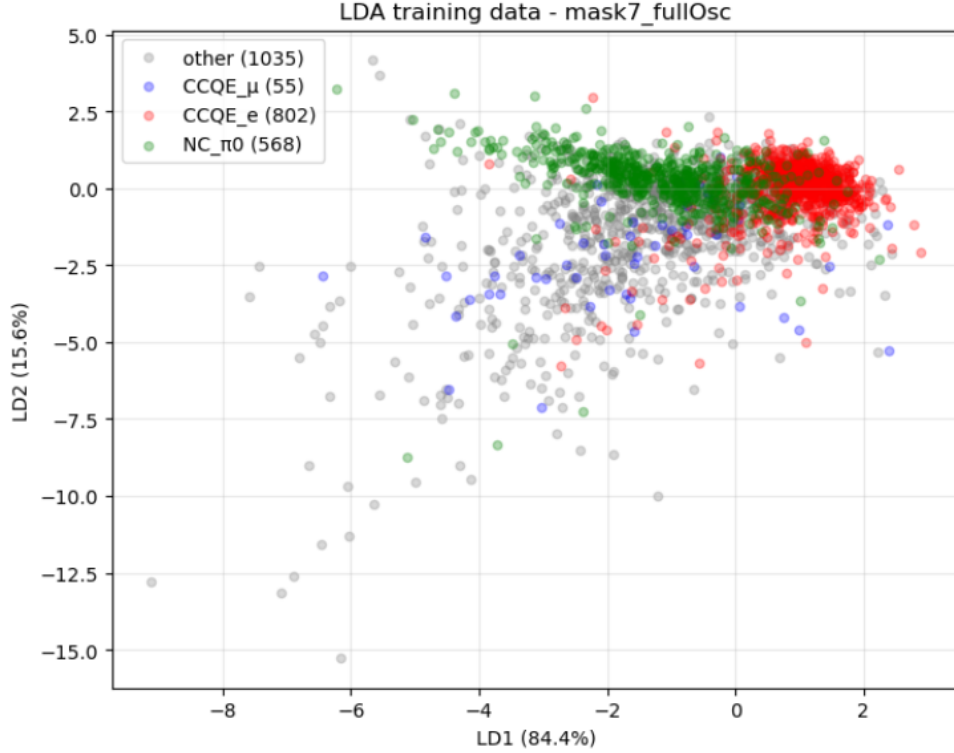
### The Nine Selection Cuts

The MiniBooNE analysis employs nine sequential selection cuts on reconstructed variables to isolate electron neutrino events: (1) exactly one subevent, (2) veto hit threshold to reject cosmic rays, (3) minimum main tank hits for reconstruction, (4) electron reconstruction radius within fiducial volume, (5) muon reconstruction endpoint away from tank walls, (6) energy range selection, (7) electron-muon likelihood ratio, (8) electron-pi-zero likelihood ratio, and (9) two-photon invariant mass cut. For the specific cut values and formulas, see Patterson 2007, pp. 165-167. These cuts progressively eliminate backgrounds while preserving signal events, with cuts 8-9 having relatively higher uncertainty than cuts 1-7.

### Our Previous Work

Professor Tanaka and I have been interested in finding the difference between Monte Carlo (MC) data and experimental data at the cluster level.

We have already experimented with a great number of physics-informed classification schemes. Unsupervised DR through Principal Component Analysis (PCA) proved unable to separate the data into meaningful clusters—the resulting 2D projections showed only a single amorphous blob regardless of how many selection cuts were applied, demonstrating that linear projection cannot capture the complex structure of the data. We thus progressed to supervised DR techniques, primarily Linear Discriminant Analysis (LDA), which proved capable of separating MC data somewhat meaningfully for some labeling schemes. We obtained the cleanest signal-to-background separation through the following categories: CCQE-e, CCQE-mu, and NC- $\pi^0$ , while ignoring all other events during training. See Figure 1.



**Figure 1:** *LDA of Monte Carlo simulated data trained exclusively on CCQE-e, CCQE-mu, and NC- $\pi^0$ , with the rest of the data plotted in the latent space as "others". Please note that the analysis was run after cut 7, as cuts 8-9 have relatively higher uncertainty. Please see Patterson 2007, pp. 165-167 for additional details on selection cuts.*

However, the "Others" category remained highly heterogeneous, and our initial labeling scheme proved too naive. We subsequently developed a more sophisticated physics-informed labeling scheme with nine categories based on particle final states and Cherenkov thresholds (distinguishing between single electron events, single photon events, single and double photon  $\pi^0$  events, single muon/pion events, and events with no Cherenkov light). Please see my presentation *MiniBooNE through the Lens of Machine Learning* for complete details of this new labeling scheme.

**The fundamental problem that undermined our approaches:** Applying linear DR to non-linear data removes critical information about the natural clustering of data. Particle physics processes create complex, non-linear relationships between the measured variables that cannot be preserved through linear projections. Additionally, a significant data handling challenge exists—approximately 90% of events do not have entries for variables associated with later cuts, requiring careful treatment of missing data either through imputation (e.g., replacing empty entries with -1) or restricting analysis to cuts 1-7. Note that variables 8-9 were only computed in MC for events that passed cuts 1-7.

## Motivation

The key limitation of our previous approaches is that they compress the data too aggressively and/or fail to capture the non-linear relationships in the high-dimensional feature space. Also, the published selection-cut approach defines a rigid 9-dimensional decision boundary without considering how events naturally cluster.

This new suggested approach addresses these limitations by leveraging neural networks to learn the non-linear structure of the data, then performing dimensionality reduction on the learned representations. Specifically, the next researcher should:

1. Train a neural network classifier on the 9 selection variables to distinguish between physics-informed event categories, implementing statistical weighting in the loss function to account for category imbalance
2. Extract activation vectors from the penultimate or final layer (after the last non-linearity)
3. Apply dimensionality reduction (PCA or t-SNE) to these activation vectors to create a 2D or 3D latent space
4. Analyze distributional shifts between MC and experimental data in this latent space

The crucial insight is that the neural network’s final layers contain representations that linearly map to the classification output. Therefore, DR at this stage preserves the discriminative information learned by the network while allowing visualization and analysis of the natural clustering of events. This approach has never been applied to MiniBooNE data and could reveal systematic differences between simulation and experiment that are invisible to traditional analyses.

By comparing the distributions of MC and experimental data in this learned latent space—examining cluster locations, densities, and separations—one can identify whether the Low-Energy Excess corresponds to a well-defined signal cluster or indicates systematic mismodeling in the simulation. Crucially, when examining distributional shifts in the principal components, one can pinpoint exactly where in the latent space the excess lies at the event level and determine whether the excess is localized in one cluster or distributed across multiple regions.

This cluster-level analysis provides a fundamentally different perspective from the published analysis based on successive selection cuts. This is exciting. Future work will require access to complete IPFS dumps and experimental data to implement the physics-informed labeling scheme and should explore various neural network architectures (varying numbers of neurons and layers, different activation functions) to find configurations with low validation loss while avoiding overfitting. For additional details, please see my slideshow *MiniBooNE through the Lens of Machine Learning*.

## References

- Patterson, R. B. (2007). *A Search for Muon Neutrino to Electron Neutrino Oscillations at  $\Delta m^2 > 0.1 \text{ eV}^2$* . PhD dissertation, Princeton University.
- Vilhelmsson Emneby, K. (2025). *MiniBooNE through the Lens of Machine Learning*. Presentation, SLAC National Accelerator Laboratory.

Sincerely

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